

Lec 03: Network design (wrap) +
Performance modelling of computer
networks

14/8/23

Recap.

- Connection-oriented service

- 3 phases

= synchronous
- reliable delivery

- Connectionless service

- Unreliable

- Communication

* Broadcast

1: all
• local network

* Unicast

1:1

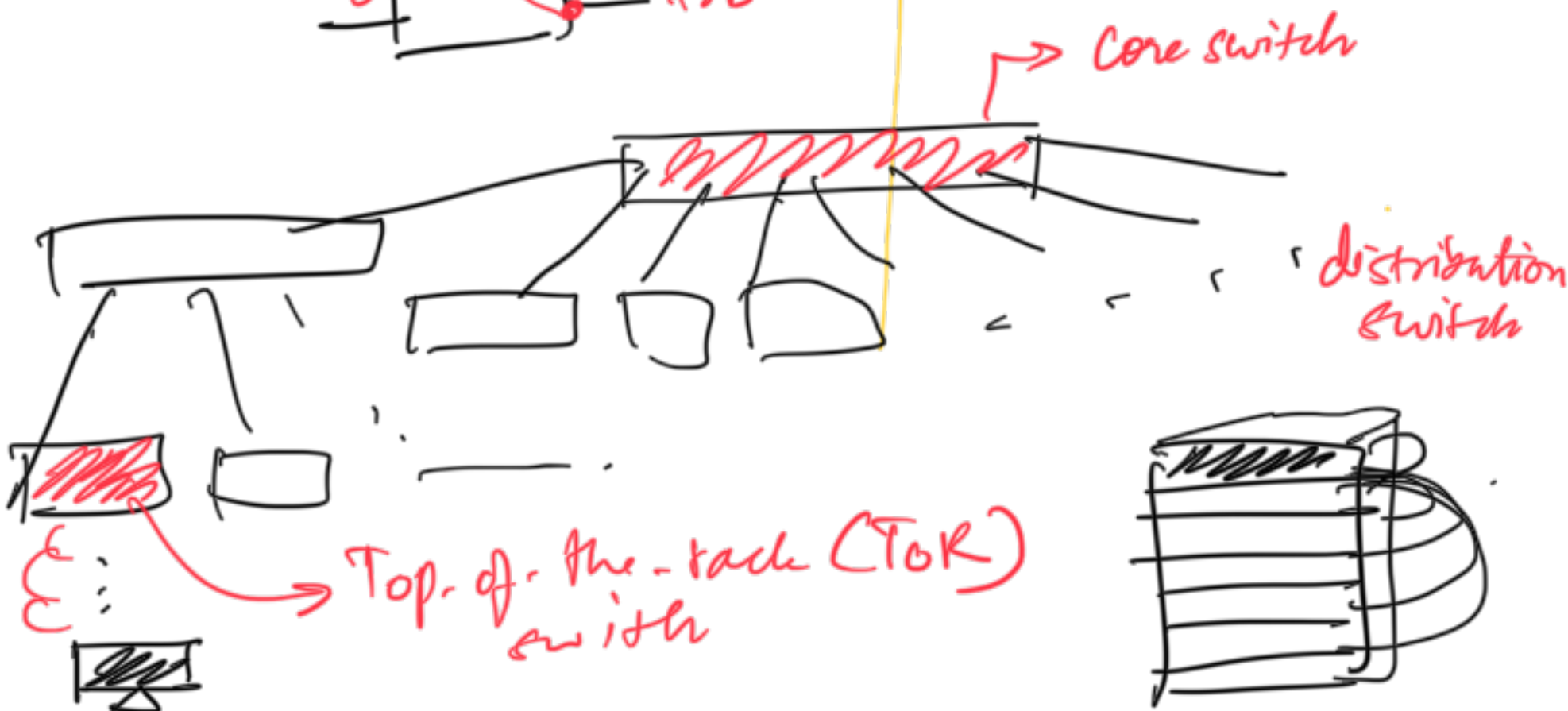
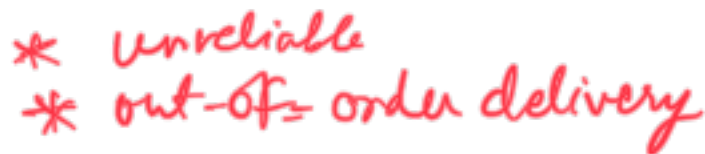
* Multicast

1: group
(many)

* Anycast

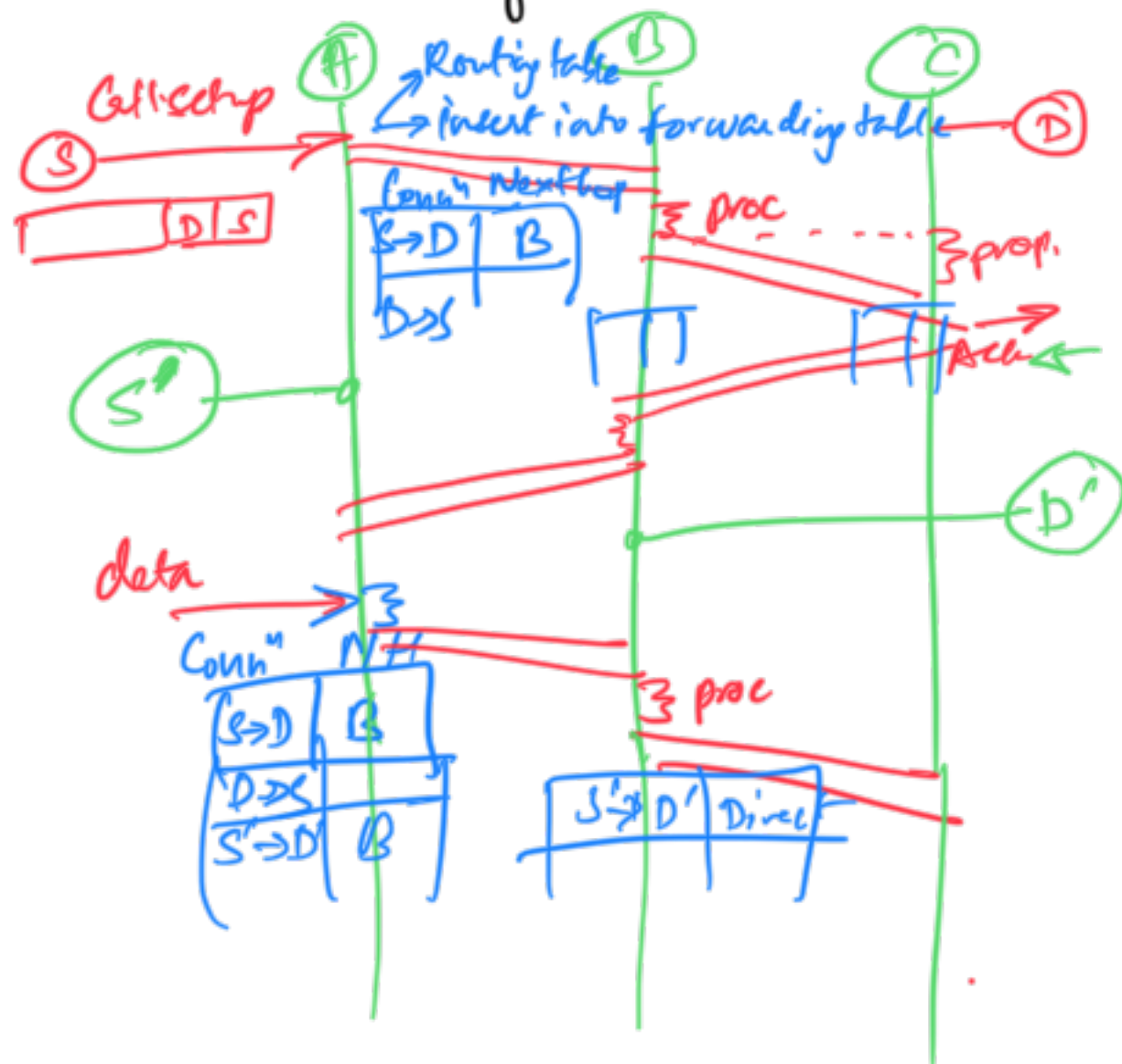
- Switching techniques

- Fixed path
Dedicated"



Virtual circuit switching $\Rightarrow$ Packet switching

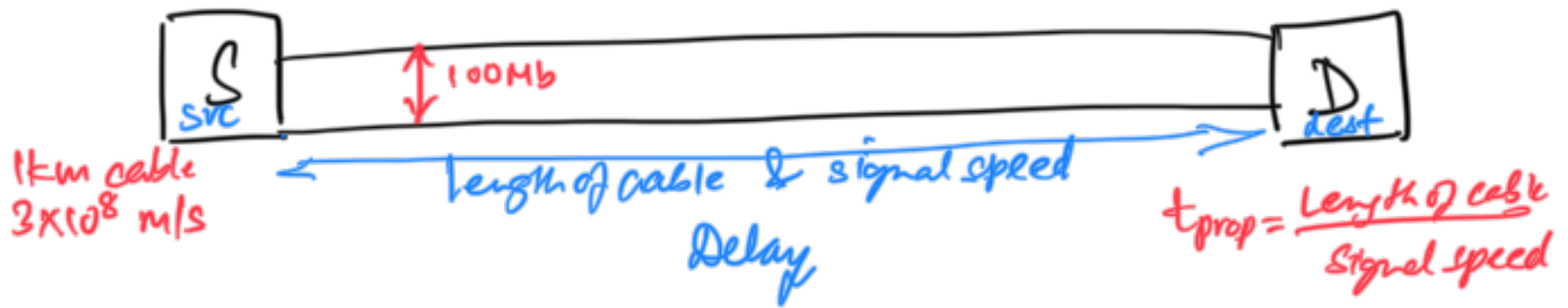
fixed path



Lec 03: Perf modelling of computer networks.

Two main attributes to measure network performance

$$\text{Throughput / Bandwidth} = \frac{\text{No. of bits that can be sent}}{\text{Total time reqd to send}}$$



t_{prop} : Time reqd for '1' bit to travel from one end to another